A counterexample to the six-point case of rank augmentation

Human Compression project

Abstract

We give an explicit rank-three string C-marking of S_{18} for which splitting one transposition from the middle generator satisfies case (3) of Theorem 4.1 in the author edition of De Saedeleer–Leemans–Mulpas, but does not give a rank-four C-marking. Generation of S_{18} follows from a tree of explicitly conjugate transpositions, and the original intersection property follows from two dihedral groups. A word in the last three new marks equals the first new mark, exhibiting the failure. This does not address cases (1) and (2), the source's small-degree census, or a complete replacement of that paper.

1 The exact statement being tested

The source [DLM] is the complete nineteen-page author edition arXiv:2212.12563v1, 23 December 2022. Its Theorem 4.1 begins on physical page 5 and continues on page 6. In the notation used here, it starts with a rank-three string C-marking (a, q, c) of the natural symmetric group S_n, $n \geq 5$, with q not a transposition. One writes $q = tb$, where t is one of its transpositions and neither endpoint of t is moved by c. The proposed new marking is (t, a, b, c). Its subgroup $\langle a, b\rangle$ is required to have all orbits of size at most three. Case (3) then assumes that $\langle t, a, b\rangle$ has an orbit of size six. Unlike cases (1) and (2), it imposes no imprimitivity or parity condition.

We use right actions and multiply from left to right; thus $x^g = g^{-1}xg$ sends the support of a transposition x to its image under g. A string C-marking consists of distinct nonidentity involutions, commuting at nonadjacent positions, with

$$\langle s_i : i \in I\rangle \cap \langle s_i : i \in J\rangle = \langle s_i : i \in I \cap J\rangle$$

for every pair of subsets of the index set. We use the standard end criterion, stated as [DLM, Proposition 5.1] and attributed there to McMullen–Schulte, Proposition 2E16(a). For a triple, it says that the intersection of the two rank-two end groups must be the common cyclic subgroup; the two distinct involutions in each end already form a rank-two C-marking.

2 The original marking

On $\{1, \dots, 18\}$ define

$$\begin{aligned} a &= (1\,2)(5\,6)(7\,8)(11\,12)(13\,14)(17\,18),\\ b &= (2\,3)(4\,5)(7\,9)(10\,11)(14\,15)(16\,17),\\ c &= (3\,4)(5\,7)(6\,8)(9\,10)(11\,13)(12\,14)(15\,16),\\ t &= (1\,18), \qquad q = tb. \end{aligned}$$

The three marks a, q, c are distinct nonidentity involutions, and a commutes with c: their common nontrivial supports are the two squares $\{5, 6, 7, 8\}$ and $\{11, 12, 13, 14\}$. The remaining transpositions of a and c are disjoint.

Lemma 1. *The group $\langle a, q, c\rangle$ is S_{18}.*

Proof. A direct calculation gives

$$aqcq = (1\,5\,8\,11\,15\,12\,7\,6\,2)(4\,9)(10\,13\,17\,18\,14).$$

Consequently $z = (aqcq)^{45} = (4\,9)$. The following table lists seventeen conjugates of this one transposition. The word in the first column is g; the second column is z^g.

g	z^g	g	z^g
1	(4 9)	q	(5 7)
c	(3 10)	qa	(6 8)
cq	(2 11)	cqa	(1 12)
cqc	(2 13)	$cqaq$	(12 18)
$cqac$	(1 14)	$cqcq$	(3 13)
$cqaqa$	(11 17)	$cqacq$	(15 18)
$cqcqa$	(3 14)	$cqcqc$	(4 11)
$cqaqaq$	(10 16)	$cqcqcq$	(5 10)
$cqcqcqa$	(6 10)		

The seventeen displayed edges form a connected graph on all eighteen points. Transpositions on the edges of a connected graph generate its full symmetric group: along a path, conjugating the next edge transposition by those already obtained gives a transposition from the initial vertex to the next vertex. Those transpositions generate all transpositions. Every displayed transposition belongs to $\langle a, q, c\rangle$, so that group contains S_{18} and is therefore equal to it. □

Lemma 2. *The original triple* (a, q, c) *is an irreducible string C-marking, of type* $[6, 90]$.

Proof. Put $r = aq$. Direct multiplication gives

$$r = (1\,3\,2\,18\,16\,17)(4\,5\,6)(7\,8\,9)(10\,11\,12)(13\,15\,14),$$

so r has order six and $r^3 = (1\,18)(2\,17)(3\,16)$. Also

$$bc = (2\,4\,7\,10\,13\,11\,9\,5\,3)(6\,8)(12\,14\,16\,17\,15).$$

The disjoint transposition t commutes with bc, so $qc = tbc$ has order ninety. The two end groups $A = \langle a, q\rangle$ and $C = \langle q, c\rangle$ are therefore dihedral, of orders twelve and one hundred eighty.

The group C preserves each of the following four sets:

$$\{1, 18\}, \quad \{2, 3, 4, 5, 7, 9, 10, 11, 13\}, \quad \{6, 8\}, \quad \{12, 14, 15, 16, 17\}.$$

Every element of A is r^i or qr^i. Since $q \in C$, any intersection larger than $\langle q\rangle$ would give a nonidentity rotation $r^i \in C$. Preservation of $\{1, 18\}$ by that rotation forces $i = 3$ modulo six. But r^3 sends 2 to 17, between two of the displayed C-invariant sets. Thus $A \cap C = \langle q\rangle$. The end criterion proves the full intersection property. The adjacent orders six and ninety also give irreducibility. □

3 The source hypotheses and the failure

The middle involution q is the product of the seven disjoint transpositions in t and b, so it is not itself a transposition. Replacing its edge $\{1, 18\}$ by the new first label yields exactly (t, a, b, c). Neither 1 nor 18 is moved by c, so the chosen edge is not adjacent to any last-label edge. The four new elements are distinct nonidentity involutions. In addition to $ac = ca$, both $tb = bt$ and $tc = ct$ hold, so they form a string marking. They generate S_{18} since they generate a, q, c. In particular, both the original and the new group actions are transitive and primitive in the natural degree eighteen. No fracture graph or additional primitivity hypothesis appears in the statement being tested.

The six orbits of $B = \langle a, b \rangle$ are

$$\{1, 2, 3\},\ \{4, 5, 6\},\ \{7, 8, 9\},\ \{10, 11, 12\},\ \{13, 14, 15\},\ \{16, 17, 18\}.$$

Each is the connected path of one a-edge and one b-edge. The subgroup $H = \langle t, a, b \rangle$ joins the first and last of these triples and fixes their union setwise. That union is connected under t, a, b, so it is one orbit of size six. The remaining four orbits still have size three. Thus both orbit hypotheses of [DLM, Theorem 4.1(3)] hold literally. Even irreducibility of the proposed new marking holds, since its adjacent orders are $4, 3, 90$.

Nevertheless its intersection property fails. The displayed cycle form of bc gives $(bc)^{45} = (6\,8)$. Set

$$w = a(bc)^2ba(cb)^8a.$$

Multiplying the given permutations shows $6^w = 1$ and $8^w = 18$. Consequently

$$((bc)^{45})^w = (1\,18) = t \in \langle a, b, c \rangle. \tag{1}$$

The index sets of the first mark t and the last three marks are disjoint, but their generated subgroups have the nonidentity element t in common. This is an explicit violation of the defining intersection property.

Theorem 3. *Theorem 4.1(3) of [DLM], in the exact author edition specified above, is false as stated. The displayed eighteen-point example satisfies all of its hypotheses and fails its conclusion.*

Proof. Lemmas 1 and 2 verify the entire input marking. The preceding orbit and edge checks verify the additional hypotheses. Equation (1) disproves the conclusion. □

The last paragraph of the source proof on physical page 16 uses the cube structure of the six-point action to conclude that the two end groups intersect only in B. That local dependence is insufficient here. Indeed, (1) implies $q = tb \in \langle a, b, c \rangle$; together with Lemma 1 it follows that $\langle a, b, c \rangle = S_{18}$. Thus $H \cap \langle a, b, c \rangle = H$, which is strictly larger than B. No disproof of cases (1) or (2), no complete source recovery, and no new conclusion about the distinct high-rank stability theorem is claimed.

4 Verification and attribution

All ingredients above reduce to finite permutation identities, the connected-graph transposition argument, and the explicitly retained end criterion. The complete program `check_case.py` verifies these identities directly on eighteen-entry permutation tuples. It enumerates only the proper subgroups of the old rank-three marking, of order at most 180, and checks all sixty-four old parabolic pairs. It never enumerates S_{18}: the seventeen-edge certificate proves that group identification. It also checks the six middle orbits, the six-point end orbit, the splitting conditions, and the explicit word (1).

The source theorem and intended local cube argument are due to De Saedeleer, Leemans and Mulpas. The elementary end criterion is retained from McMullen–Schulte as cited by that source. The complete written proof and a separate Lean supplement have passed distinct review. The supplement verifies the actual output obstruction; the input hypotheses remain written proofs. There is no end-to-end Lean certificate of the source theorem, and no numerical compression claim.

References

[1] J. De Saedeleer, D. Leemans and J. Mulpas, *A rank augmentation theorem for rank 3 string C-group representations of the symmetric groups*, arXiv:2212.12563v1, 23 December 2022, Theorem 4.1 and §5.3, physical pages 5–6 and 14–16.